```
from games4e import query_player, mcts_player, random_player
from utils4e import MCT_Node, ucb
```

Interactive Connect Four Game

Play Connect Four as the query player (X) against the heuristic alpha-beta search player (O).

```
# Define query_player compatible with our notebook's API
def query_player_notebook(game, state):
    """Make a move by querying standard input."""
    print("\ncurrent state:")
    game.display(state)
    print("available moves: {}".format(game.actions(state)))
    print("")
    move = None
    if game.actions(state):
        move_string = input('Your move? ')
        try:
            move = eval(move_string)
        except NameError:
            move = move_string
    else:
        print('no legal moves: passing turn to next player')
    return move

# Create Connect Four game and play
# You are X (query player), AI is O (heuristic alpha-beta)
print("Connect Four: You (X) vs AI (O)")
print("Enter your move as a tuple, e.g., (0, 5) for column 0")
print("Remember: In Connect Four, you can only play in the lowest empty
square of each column")
print("\nLet's play!\n")

result = play_game(ConnectFour(),
                   dict(X=query_player_notebook,
O=player(h_alphabeta_search)),
                   verbose=True)

print("\n" + "="*50)
if result.utility == 1:
    print("X (You) WINS! 🎉")
elif result.utility == -1:
    print("O (AI) WINS! 🤖")
else:
```

```
    print("It's a DRAW! 🤝")
print("="*50)
```

```
[Interactive game output - execution specific]
```

MCTS vs Heuristic Alpha-Beta Search

Now let's test Monte Carlo Tree Search against the heuristic alpha-beta search player over 5 rounds.

```
# Import MCTS player from games4e
from games4e import monte_carlo_tree_search

# Create MCTS player wrapper
def mcts_player(game, state):
    """MCTS player that returns just the move (not the full action)."""
    return monte_carlo_tree_search(state, game, N=1000)

# Run 5 rounds: MCTS (X) vs Heuristic Alpha-Beta (O)
print("Running 5 rounds: MCTS (X) vs Heuristic Alpha-Beta (O)")
print("="*60)

results = {'X': 0, 'O': 0, 'Draw': 0}

for round_num in range(1, 6):
    print(f"\n{'='*60}")
    print(f"ROUND {round_num}")
    print(f"{'='*60}")

    result = play_game(ConnectFour(),
                       dict(X=mcts_player, O=player(h_alphabeta_search)),
                       verbose=False)

    if result.utility == 1:
        winner = 'X (MCTS)'
        results['X'] += 1
    elif result.utility == -1:
        winner = 'O (Alpha-Beta)'
        results['O'] += 1
    else:
        winner = 'Draw'
        results['Draw'] += 1

    print(f"\nRound {round_num} Result: {winner}")
    print(result)

# Final summary
```

```
print("\n" + "="*60)
print("FINAL RESULTS AFTER 5 ROUNDS")
print("="*60)
print(f"MCTS (X) wins:              {results['X']}")
print(f"Alpha-Beta (O) wins:        {results['O']}")
print(f"Draws:                      {results['Draw']}")
print("="*60)

if results['X'] > results['O']:
    print("🏆 WINNER: Monte Carlo Tree Search (MCTS)")
elif results['O'] > results['X']:
    print("🏆 WINNER: Heuristic Alpha-Beta Search")
else:
    print("🤝 Overall: TIE")
print("="*60)
```

```
Running 5 rounds: MCTS (X) vs Heuristic Alpha-Beta (O)
============================================================

============================================================
ROUND 1
============================================================

Round 1 Result: X (MCTS)
. . . . . . .
. . . O . . X
. . . O . X O
. X . X X O X
. O . X O X X
O O . X X O O


============================================================
ROUND 2
============================================================

Round 2 Result: O (Alpha-Beta)
```

```
. X O . X . O
O O O . X O O
O O X . O X O
X X X . X X X
O O O O O X X
X O X X X O X


============================================================
ROUND 3
============================================================

Round 3 Result: X (MCTS)
. X . X O . O
. O . O O . X
. X . O X X O
. X X X O O X
O X O X O X X
X O O X X O O


============================================================
ROUND 4
============================================================

Round 4 Result: X (MCTS)
. X O X . O O
. O X O . O O
. O X O . X O
X X X X . X X
X X O X O X O
X O X X O O O


============================================================
ROUND 5
============================================================
```

```
Round 5 Result: X (MCTS)
. . . . . . .
. . . . . . .
. . . O . . .
. X X X X . .
. O O X O . .
. O X X O X O



============================================================
FINAL RESULTS AFTER 5 ROUNDS
============================================================
MCTS (X) wins:              4
Alpha-Beta (O) wins:        1
Draws:                      0
============================================================
🏆 WINNER: Monte Carlo Tree Search (MCTS)
============================================================
```